

<<P15 Approach to a Patient with Respiratory Disease _ Preventative Care (Pathophysiology)
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SESSION OBJECTIVES



Learn how to take a medical history of patients with lung disease



Understand pathophysiology of presenting symptoms

Dyspnea, Cough, Chest Pain, Hemoptysis



Know the technique and interpretation of the findings on PE of the lung and thorax



Know the basic pulmonary diagnostic techniques → invasive and non-invasive



Understand High Risk behavior for lung disease

Smoking, Vaping, and Lifestyle Modifications



Review Vaccine Preventable Respiratory Illness

Influenza, COVID-19, RSV, Pertussis, Pneumococcus



Review additional Preventable Diseases

Identification and Treatment of Latent Tuberculosis

Risk factors for venous thromboembolism, and the rationale and approach for prevention

OVERVIEW



Taking a medical history of patients with lung disease

Pathophysiology of Presenting Symptoms

Technique & Interpretation of the lung and thorax on PE

Basic pulmonary diagnostic techniques

Understand High Risk behavior for lung disease

Review Vaccine Preventable Respiratory Illness

Review additional Preventable Diseases

Case Studies

Case 1

23-year-old college student, previously healthy, presents with fever for 4 days, associated body aches, fatigue and non-productive cough.

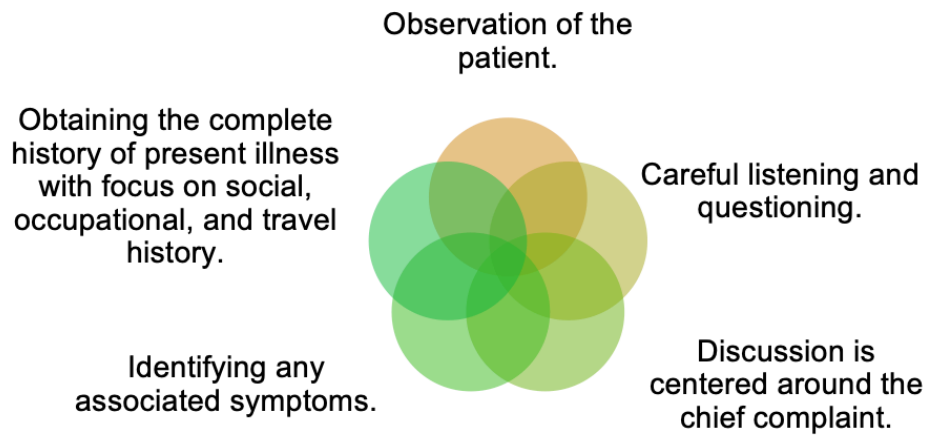
Case 2

58-year-old former smoker who quit 10 years ago, presents with three weeks of cough with scarce yellowish sputum, severe exertional dyspnea and weakness.

Case 3

13-year-old male, known history of severe eczema and environmental allergies, presents with progressive exertional dyspnea, wheeze, and nocturnal cough. Symptoms improve with rest.

Interviewing a Pulmonary Patient



OVERVIEW



Taking a medical history of patients with lung disease

Pathophysiology of Presenting Symptoms

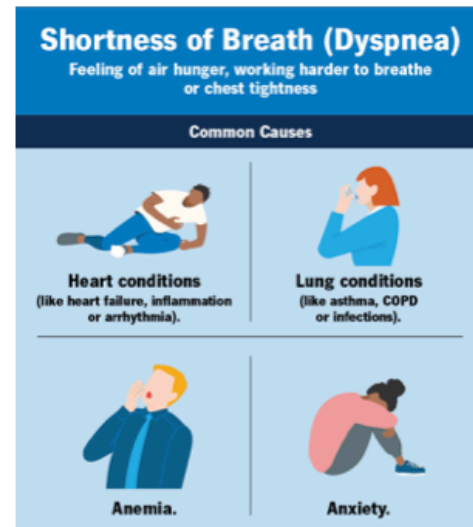
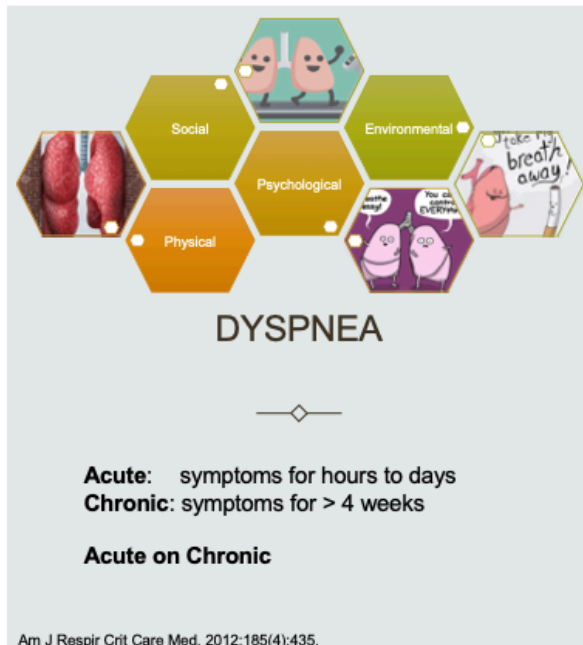
Technique & Interpretation of the lung and thorax on PE

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Deconditioning, Obesity, Neuromuscular Disease

Dyspnea can present as:

- Acute when new shortness of breath develops over a very short time
- Chronic when it has been present for more than 4 weeks
- Acute on chronic when a patient with baseline dyspnea experiences a new acute worsening on top of their usual symptoms

Conditions

- Patients with chronic lung diseases such as chronic bronchitis, COPD, or asthma often have chronic dyspnea and then present with acute on chronic episodes during exacerbations
- Cardiac problems including CHF and chronic arrhythmias such as atrial fibrillation can also manifest as shortness of breath, so dyspnea in these patients should prompt evaluation of both heart and lungs
- Systemic conditions like anemia can make patients feel short of breath because reduced oxygen carrying capacity leads to a sense of air hunger even when the lungs are structurally normal
- Anxiety can cause or amplify dyspnea, often through hyperventilation and increased awareness of breathing, and may coexist with true cardiopulmonary disease

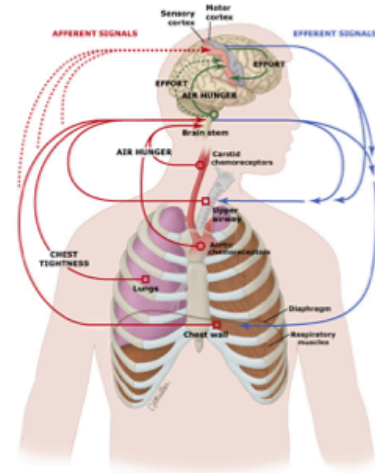
Dyspnea Pathophysiology

The respiratory system is designed to maintain homeostasis by adjusting ventilation based on acid-base status and the partial pressure of oxygen

This involves chemoreceptors and mechanoreceptors

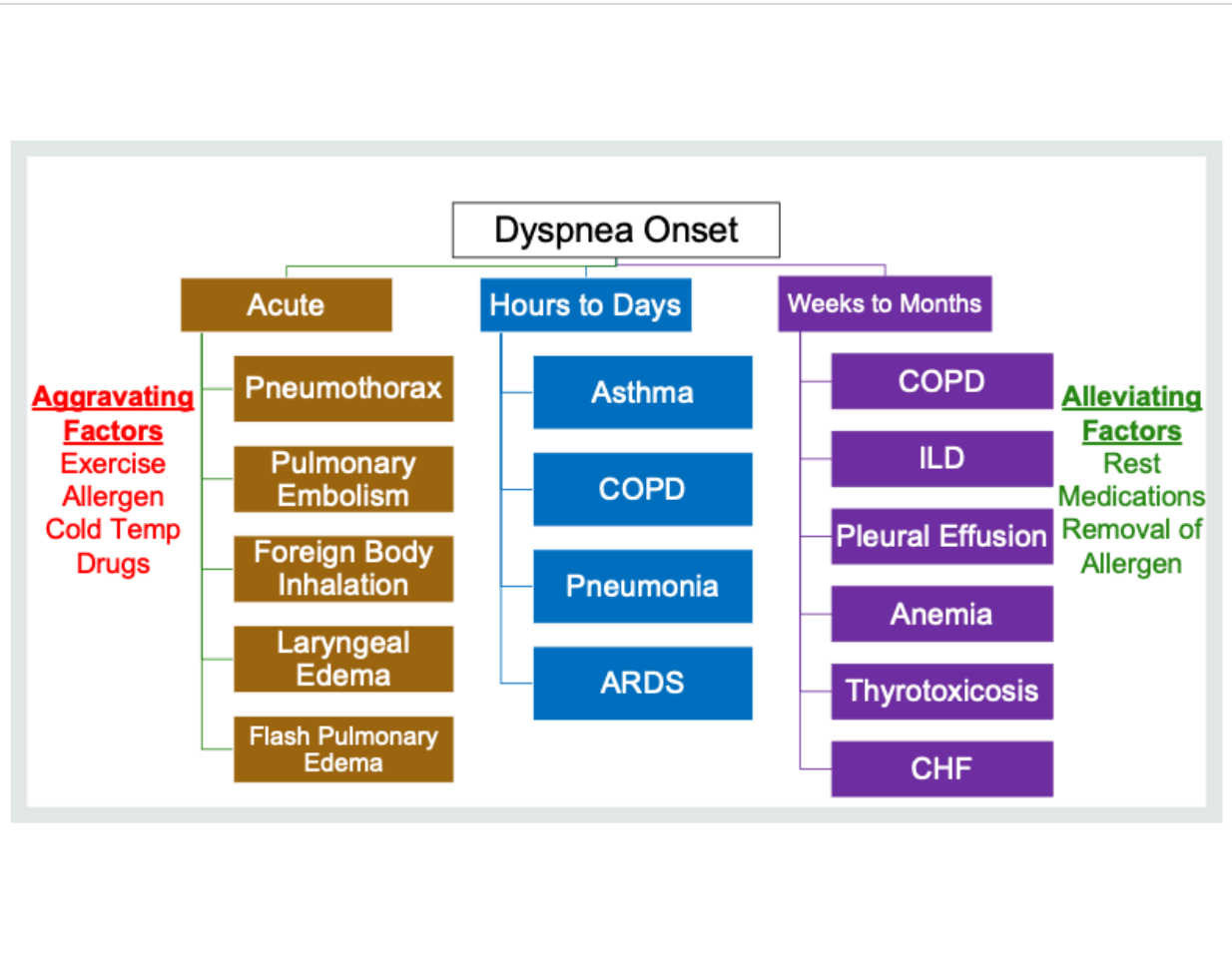
Disturbance in this balance leads to the sensation of dyspnea

Efferent and afferent signals that contribute to the sensation of dyspnea



Dyspnea arises from complex, integrated pathways that monitor and respond to changes in breathing, and multiple sensors can generate the sensation of being short of breath

- Chemoreceptors detect alterations in acid base status and partial pressures of gases, especially changes in carbon dioxide and oxygen, and then signal the brainstem to adjust ventilation
- Mechanoreceptors in the lungs, chest wall, and respiratory muscles sense the physical effort of breathing and mechanical loads on the respiratory system and send afferent feedback that shapes the perception of air hunger or chest tightness
- Any disturbance in these chemical or mechanical inputs can produce dyspnea, even when lung structure is normal, because the brain interprets mismatches between expected and actual respiratory status as shortness of breath
- Anxiety with hyperventilation is a common example, where excessive ventilation lowers carbon dioxide, activates chemoreceptors, and drives a feedback loop that amplifies the subjective sensation of dyspnea despite normal oxygenation and lung mechanics



When a patient reports 1 day of new dyspnea, you should prioritize acute life threatening causes such as pneumothorax, pulmonary embolism, inhaled foreign body, laryngeal edema from anaphylaxis, or acute pulmonary edema

- Subacute dyspnea developing over days often reflects progression of an underlying pulmonary disease such as pneumonia or a flare of chronic lung disease
- For more chronic dyspnea you must take a detailed HPI and specifically ask what makes symptoms better or worse because exertional dyspnea is typical of chronic pulmonary and cardiac conditions

Other Considerations

Exertional dyspnea

- Seen in Asthma, COPD, CHF

Paroxysmal Nocturnal Dyspnea

- SOB that awakens a patient, often after 1 or 2 hours of sleep, often relieved in upright position
- More common in CHF

Positional:

- Bendopnea (SOB shortly after bending at the waist) – CHF, Obesity
- Orthopnea (SOB while lying flat and relieved by sitting or standing) - CHF, Obesity, inspiratory muscle weakness

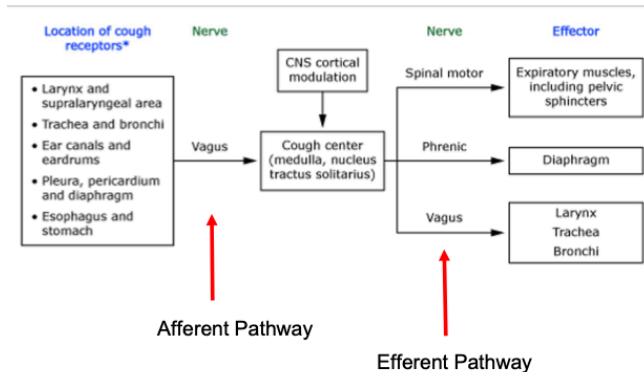
Paroxysmal nocturnal dyspnea is dyspnea that awakens a patient after they have fallen asleep and is relieved by sitting upright or even getting up to open a window and is classically seen in CHF

- Positional patterns help localize the problem because bendopnea, where patients feel short of breath when bending at the waist to tie their shoes, can occur when increased abdominal girth or CHF pushes the diaphragm upward and compresses the lungs
- Orthopnea is dyspnea that occurs when lying flat awake and improves with sitting or standing and is commonly due to CHF or obesity, where abdominal contents mechanically impair diaphragmatic excursion, or to inspiratory muscle weakness in neuromuscular disease
- If a patient with orthopnea does not appear obese and has no obvious CHF, you should suspect neuromuscular weakness and compare pulmonary function testing in upright versus supine positions, expecting worse pulmonary function when supine if a neuromuscular disorder is present

COUGH

Reflex act of forceful expiration against a closed glottis generating positive intrathoracic pressure as high as 300 mmHg.

The point is to clear the airway from whatever is triggering it.

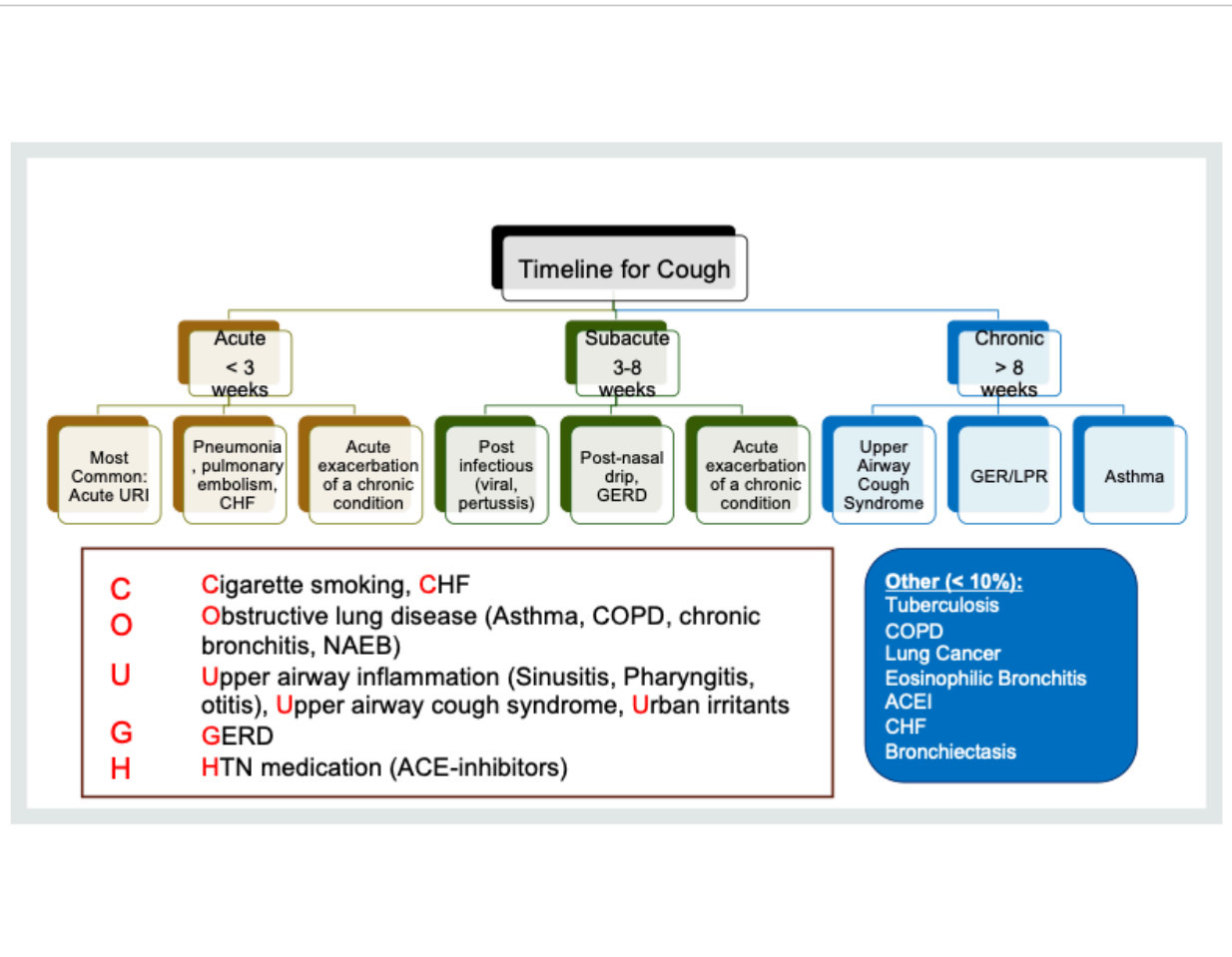


Cough Reflex Arc

- Each cough starts by stimulation of this path
- Cough receptors get stimulated.
- These are mechanical (touch, displacement) and chemical (acid, cold, heat, irritants).

Cough is a powerful reflex that begins when a stimulus activates cough receptors, which send afferent signals to the central cough center and then through efferent pathways to respiratory muscles to generate a forceful expiratory effort

- A single cough can create positive intrathoracic pressures up to 300 mm Hg, which explains complications such as rib fractures, muscle strain, and pulled intercostal muscles in patients with severe coughing



Clinically, cough is categorized by duration into acute, subacute, and chronic, and the likely etiologies change across this timeline

- Acute cough is most often due to a URI, whereas chronic cough is commonly caused by untreated GERD, upper airway cough syndrome, or undiagnosed or poorly controlled asthma
- Subacute cough often represents conditions that have not yet fully declared themselves, such as allergic rhinitis with significant postnasal drip
- **ACE inhibitors are a classic medication cause of chronic cough, and the cough can start at any time during therapy, even years after initiation, which is a frequent test question trap**

Other Considerations

Nocturnal

- Postnasal drip
- GERD
- Chronic bronchitis
- Asthma
- Aspiration
- OSA
- CHF



Sputum Production

Nocturnal cough is especially associated with postnasal drip, because secretions pool in the back of the throat when supine and overstimulate pharyngeal receptors, leading to prolonged nighttime coughing in patients with URIs or allergic rhinitis

- GERD is another major cause of nocturnal cough, since relaxation of the lower esophageal sphincter during sleep allows gastric acid to reach the upper airway and trigger cough receptors in the larynx and pharynx

Sputum

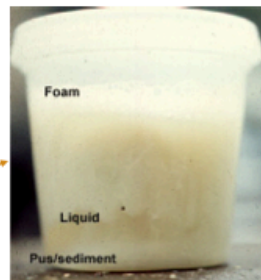
Considerations: Amount, Consistency, Color, Postural variation, Smell

Amount

Copious Amount

- Bronchiectasis
- Lung Abscess
- Necrotizing pneumonia
- Alveolar cell carcinoma
- Empyema rupturing into bronchus

(Bronchorrhoea - >100ml sputum/day)



Consistency

- Serous - Upper Respiratory tract Infection, Bronchoalveolar carcinoma
- Muroid - Chronic bronchitis, Bronchial Asthma
- Mucopurulent - Bacterial infection

When evaluating cough you must ask whether it is productive and then characterize the sputum in detail, because amount, consistency, color, and smell narrow the differential

- Very large volumes of sputum suggest serious pathology such as lung abscess, necrotizing pneumonia, or chronic bronchiectasis where patients continuously overproduce mucus
- Muroid sputum is typical of chronic bronchitis, whereas visible pus mixed with mucus suggests an acute bacterial infection

Sputum

Color of sputum

Yellow / Green

Black

Pink & Frothy

Rusty

"Red currant jelly"

Blood tinged / streaking

"Anchovy sauce"

Bacterial Infection
Coal worker pneumoconiosis
Pulmonary edema
Pneumococcal pneumonia
Klebsiella pneumonia
Tuberculosis pneumonia
Ruptured amoebic liver abscess

Postural variation

- Lung abscess
- Bronchiectasis

Foul Smell

- Lung abscess
- Bronchiectasis
- Anaerobic bacterial infection



Patients who describe blood tinged sputum may have tuberculosis pneumonia, and anchovy sauce sputum is linked to a ruptured amoebic liver abscess communicating with the bronchial tree

- You should also ask about odor because copious foul smelling sputum strongly suggests bronchiectasis, lung abscess, or anaerobic bacterial lung infection, often after aspiration

CHEST PAIN



Site

Onset

Duration

Severity

Character

Radiation

Associated Symptoms

Aggravating / Relieving Factors

Diurnal / Seasonal Variation

Retrosternal Pain

Chest Pain - Causes

Upper

- Tracheitis

Mid and Lower

- Mediastinitis
- Mediastinal Tumor
- GERD
- Achalasia cardia

Diffuse Esophageal Spasm

Angina

Pleural Inflammation

- Catchy pain, ↑ deep inspiration, stabbing in character

Pancoast Tumor

- Shoulder/Arm pain d/t compression C8
- T1-2 roots in sharp shooting pain along course of nerve

Erosion of ribs

- Constant dull aching chest pain

Tietze's syndrome

- Costochondritis, unknown etiology

Chest pain can be emergent from cardiac causes or noncardiac yet still serious from pulmonary or pleural disease, so history is critical to distinguish them

- You should use a detailed HPI to localize and characterize the pain because patients' descriptions often map to specific pathologies
- Pleuritic chest pain is typically sharp and worsens with deep inspiration, which suggests inflammation or irritation of the pleura rather than primary myocardial ischemia
- A classic pulmonary cause of new pleuritic pain is pulmonary embolism that leads to pulmonary infarction and secondary pleural involvement, so onset of sudden inspiratory chest pain with dyspnea should prompt evaluation for PE

HEMOPTYSIS

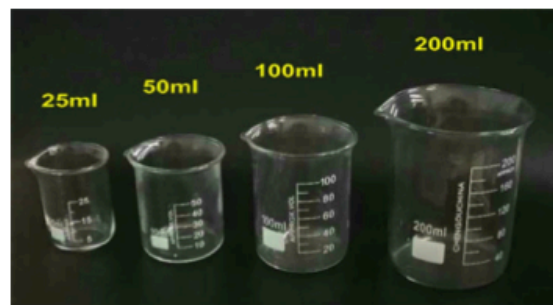


Term for coughing up blood from respiratory tract

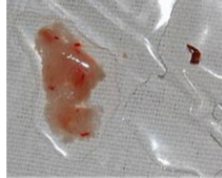
This can range from blood streaks to frank blood

Severe/life threatening hemoptysis is considered:

1. > 150ml in 24 hrs
2. > 100ml/hr
3. Clinically diagnosed when results in a life-threatening event



Hemoptysis is coughing up blood, and you must distinguish small blood tinges in mucus from coughing up frank blood, because larger volumes can be rapidly life threatening.



Hemoptysis

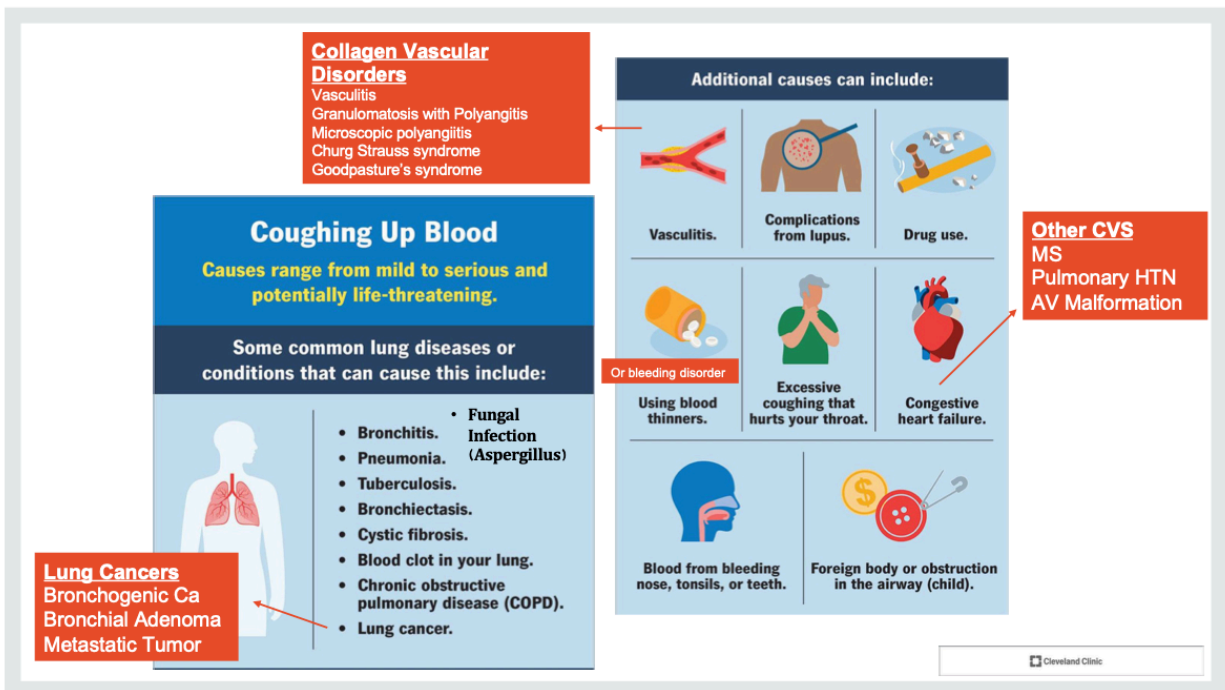


Can originate from the pulmonary arterial and bronchial arterial systems

The majority are from the PA circulation, which is a low-pressure system. These are typically considered non-life threatening

Some cases originate from BA circulation

Although the bronchial artery circulation is responsible for only 2% of the total vascular supply to the lung, bleeding from a bronchial artery is the cause of life-threatening hemoptysis in 90 percent of cases



Common pulmonary causes include infections such as bronchitis, pneumonia, tuberculosis, necrotizing infections, and bronchiectasis or cystic fibrosis, where blood is usually mixed with sputum rather than pure blood

- Other important causes are pulmonary embolism, primary or metastatic lung cancers, and inhalational injury from recreational drugs such as crack cocaine
- Systemic and vascular contributors include excess anticoagulation when “blood is too thin,” AV malformations, mitral stenosis, and pulmonary hypertension, as well as vasculitic collagen vascular diseases that can flare with pulmonary capillaritis and bleeding
- In children and young adults you must also consider foreign body aspiration with local trauma as a cause of hemoptysis
- Finally, always confirm that the blood is truly from the lower respiratory tract, because posterior epistaxis or bleeding from the upper airway can be misinterpreted by patients as hemoptysis when the source is actually above the lungs

Causes of Hemoptysis

Mn. BATTLE CAMP

B
A
T
T
L
E
C
A
A
M
P

Bronchitis, Bronchiectasis
Aspergilloma
Tumor
Tuberculosis
Lung abscess
Emboli (Pulmonary)
Coagulopathy
Autoimmune disorders
AV malformation
Alveolar haemorrhage
Mitral stenosis
Pneumonia

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Case Studies

Case 1

23-year-old college student, previously healthy, presents with fever for 4 days, associated body aches, fatigue and non-productive cough.

**Acute
Presentation
Cough**

Case 2

58-year-old former smoker who quit 10 years ago, presents with three weeks of cough with scarce yellowish sputum, severe exertional dyspnea and weakness.

**Subacute
Presentation
Cough, Dyspnea**

Case 3

13-year-old male, known history of severe eczema and environmental allergies, presents with progressive exertional dyspnea, wheeze, and nocturnal cough. Symptoms improve with rest.

Symptoms first began 1 year ago and are worse with Fall/Spring Season when participating in Track.

**Chronic
Presentation
Cough, Dyspnea**

OVERVIEW



Taking a medical history of patients with lung disease



Pathophysiology of Presenting Symptoms



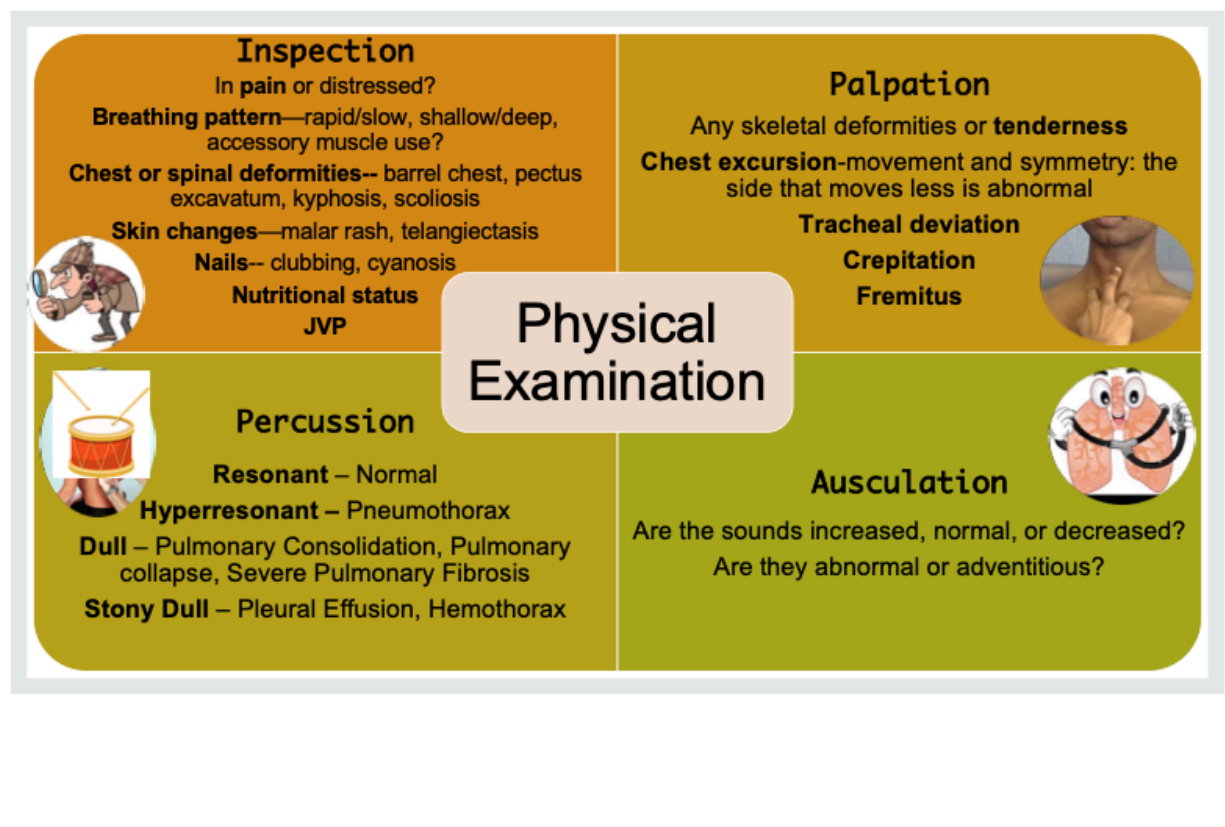
Technique & Interpretation of the lung and thorax on PE

Basic pulmonary diagnostic techniques

Understand High Risk behavior for lung disease

Review Vaccine Preventable Respiratory Illness

Review additional Preventable Diseases

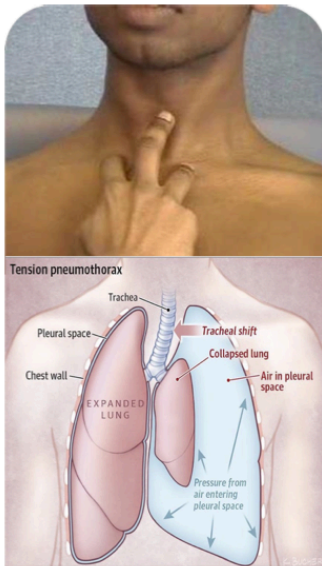


Respiratory physical examination starts with careful inspection, because simple visual cues can reveal systemic diseases that underlie lung pathology

- Noticing a malar rash on the face should prompt consideration of rheumatologic conditions such as autoimmune connective tissue disease that may be causing interstitial lung disease or pleuritis
- You must know how and where to palpate the chest to assess chest wall tenderness, chest expansion, and fremitus, because asymmetry can indicate consolidation, effusion, or obstruction on the side that moves less

Palpation

Tracheal Deviation



Ipsilateral

- ☐ Atelectasis
- ☐ Fibrosis
- ☐ Pneumectomy

Contralateral

- ☐ Pleural effusion
- ☐ Pneumothorax

Palpation

Assessing Symmetry of Chest Wall Movement and Excursion



Upper thoracic expansion



Anterior thoracic expansion



Posterior thoracic expansion

Palpation

Decreased Chest Excursion

Unilateral

- ☐ Pleural Effusion
- ☐ Empyema
- ☐ PTX
- ☐ Consolidation
- ☐ Atelectasis

Bilateral

- ☐ Asthma
- ☐ Emphysema
- ☐ Pulmonary Fibrosis

Percussion

Feel with the flat of a hand or the ulnar border of the hand

Patient is asked to say "Ninety-Nine"

Feel for Vibration of Chest Wall

☐ Result of Sound Transmitting Through Lung Tissue

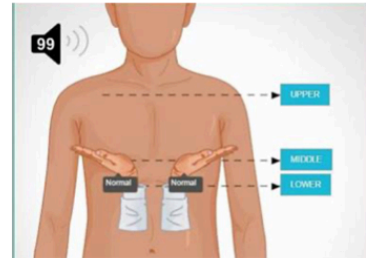
Feel for changes in Intensity of Fremitus by Palpating Chest Wall



Increased
Consolidation



Decreased
Pleural Effusion
Thick chest wall
Pneumothorax
Pleural fibrosis
Emphysema
Atelectasis



Percussion

Assess Resonance



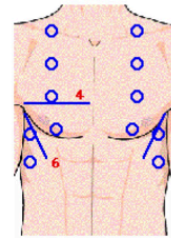
Keep your finger flat!

University of Cape Town

Percussion note	
Type	Detected over
Resonant	Normal lung
Hyperresonant	Pneumothorax
Dull	Pulmonary consolidation
	Pulmonary collapse
	Severe pulmonary fibrosis
Stony dull	Pleural effusion
	Haemothorax

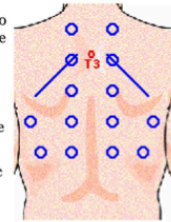
Percussion: Anterior Chest

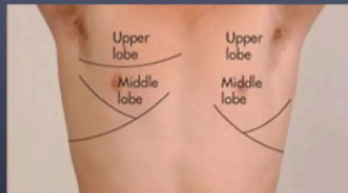
1. Percuss from side to side and top to bottom using the pattern shown in the illustration.
2. Compare one side to the other looking for asymmetry.
3. Note the location and quality of the percussion sounds you hear.



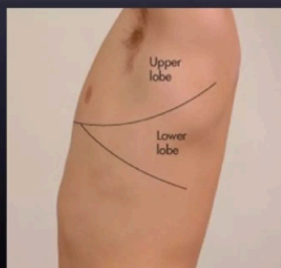
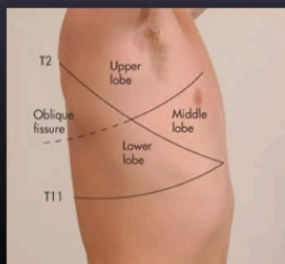
Percussion: Posterior Chest

1. Percuss from side to side and top to bottom using this pattern. Omit the areas covered by the scapulae.
2. Compare one side to the other looking for asymmetry.
3. Note the location and quality of the percussion sounds you hear.
4. Find the level of the diaphragmatic dullness on both sides.





Auscultation



Auscultation

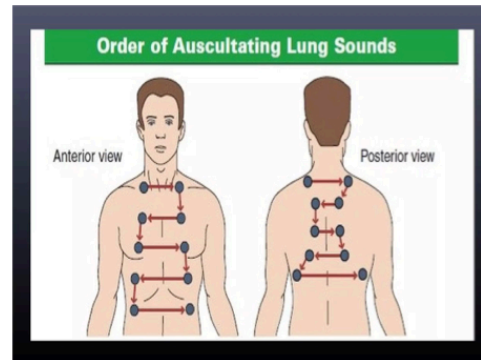
Ideally

- Patient in comfortable seated position
- Quiet setting

Auscultate the bare chest

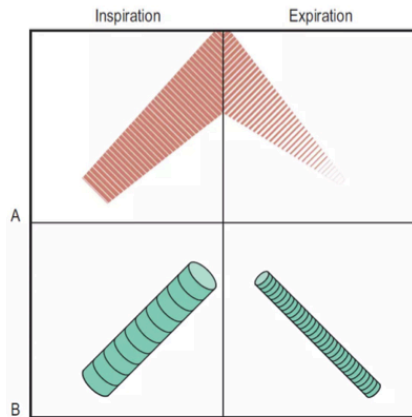
Warm stethoscope is preferred to ensure patient comfort

Auscultate symmetrically in a systematic pattern



Are the sounds increased,
normal, or decreased?
Are they abnormal or
adventitious?

Auscultation – Breath Sounds

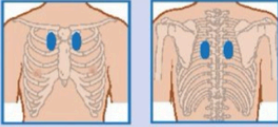
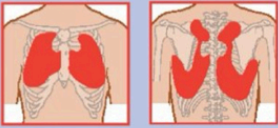


Diagrammatic representation of breath sounds.
(A) Vesicular. (B) Bronchial. Note the gap between inspiration and expiration and change in pitch and the blowing, tubular quality of bronchial breath sounds.

Breath Sounds

- Normal
 - Tracheal
 - Bronchial
 - Bronchovesicular
 - Vesicular
- Adventitious
 - Crackles (Rales)
 - Wheeze
 - Rhonchi
 - Stridor
 - Pleural Rub
- Abnormal
 - Absent/Decreased
 - Bronchial

Auscultation Breath Sounds

Table 1. Quality of normal breath sounds				
Breath sound	Intensity and pitch	Inspiratory: expiratory ratio	Positions to hear sounds	
Tracheal 	Very loud, high pitch	Inspiratory and expiratory sounds equal	Over the trachea (above the subclavicular notch)	
Bronchial 	Loud, relatively high pitch	Inspiratory sound shorter than expiratory	Over the manubrium (just above the clavicles)	
Bronchovesicular 	Medium loudness, intermediate pitch	Inspiratory and expiratory sounds equal	First and second intercostal spaces next to the sternum and between the scapula	
Vesicular 	Soft, relatively low pitch	Inspiratory sound longer than expiratory	Most of the lung field	

AUSCULTATION ADVENTITIOUS BREATH SOUNDS



ABNORMAL LUNG SOUNDS

WHEEZING 	HIGH PITCHED, CONTINUOUS MUSICAL SOUND - Air is moving through a narrowed airway - Heard on inspiration and expiration - Ex: Asthma, inflammation
CRACKLES 	HIGH PITCHED, POPPING SOUND - Air is being forced through a narrowed airway - Heard on inspiration mostly - Ex: Infection of bronchi, bronchioles, or alveoli; pulmonary edema, ARDS FINE: sounds like hair moving through fingers COARSE: low bubbling sound; clears with coughing
RHONCHI	LOW PITCHED RUMBLE (LIKE A SNORE) - Narrowed airway in trachea or bronchi - Heard on inspiration and expiration - Clears with cough - Ex: COPD, CF, bronchitis 
STRIDOR 	HIGH PITCHED WHEEZING SOUND - Obstruction of upper airway - disrupted air flow in larynx or trachea - Heard on inspiration mostly - Ex: Croup, epiglottitis, foreign body in airway
PLEURAL FRICTION RUB	HARSH GRATING, LEATHERY SOUND - Friction of Inflamed and roughened pleura - Heard on inspiration and expiration - Often mistaken for coarse crackles - Ex: Rheumatic fever, TB, autoimmune disorder, chest trauma
ABSENT OR DIMINISHED	- Increased distance between lung and chest wall - Breath sounds limited by: - Asthma, emphysema - Tumor, collapse, complete obstruction - Effusion, fibrosis 

Auscultation

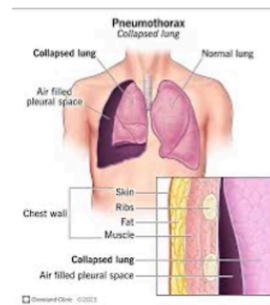
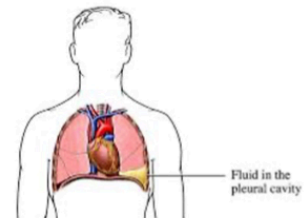
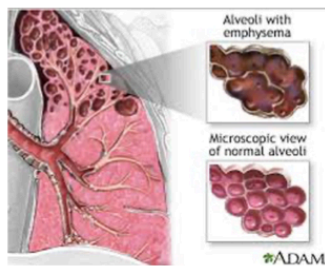
Decreased Breath Sounds

Atelectasis

Emphysema

Pleural effusion

Pneumothorax



Auscultation Vocal Sounds

- Listen for voice sounds to be heard near the earpiece of the stethoscope
- Normally vocal sounds are heard as weak, muffled and indistinct rumbling
- **Bronchophony**
 - ❑ Auscultation of symmetrical areas on patients bare chest while patient says "99"
 - ❑ Abnormal transmission of spoken phrase heard best over area of consolidation
- **Egophony**
 - ❑ voice sound has a nasal or bleating quality.
 - ❑ On saying "E" (eeeeee) it will sound like "A" (aaaaa) (E to A sign)
 - ❑ Seen in consolidation.
- **Whispered pectoriloquy**
 - ❑ Whisper "1-2-3" - at the end of expiration are heard clear and distinct.
 - ❑ Heard best over area of consolidation.



Case Studies

Case 1

23-year-old college student, previously healthy, presents with fever for 4 days, associated body aches, fatigue and non-productive cough.

Clinical Exam: Sats 96%, mild tachypnea, flushed appearance, Temp 102.3 F, clear lungs with equal/good air entry

Case 2

58-year-old former smoker who quit 10 years ago, presents with three weeks of cough with scarce yellowish sputum, severe exertional dyspnea and weakness.

Clinical Exam:
Tachypnea, bibasilar crackles, Saturations of 85%

Case 3

13-year-old male, known history of severe eczema and environmental allergies, presents with progressive exertional dyspnea, wheeze, and nocturnal cough. Symptoms improve with rest.

Clinical Exam: Sats 98%, RR 16, good air entry, diffuse scattered expiratory wheeze

OVERVIEW



Taking a medical history of patients with lung disease



Pathophysiology of Presenting Symptoms



Technique & Interpretation of the lung and thorax PE

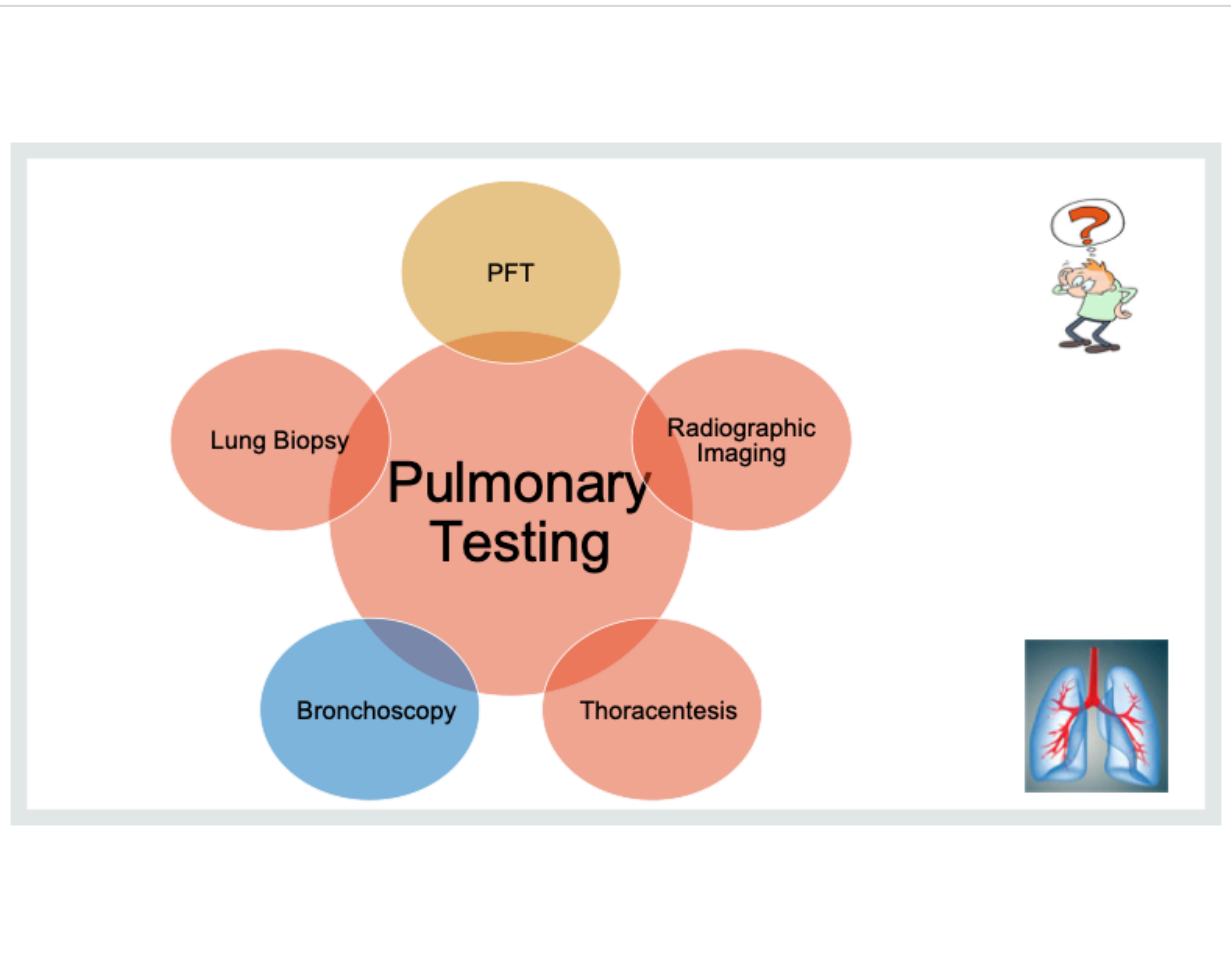


Basic pulmonary diagnostic techniques

Understand High Risk behavior for lung disease

Review Vaccine Preventable Respiratory Illness

Review additional Preventable Diseases



MRI is rarely used for primary lung evaluation because it images pulmonary parenchyma poorly, but it can be useful for other thoracic structures, so lung diagnosis often relies more on tissue sampling and physiologic testing

- Lung tissue can be obtained by biopsy directly or via bronchoscopy, which is usually an ambulatory or bedside procedure done under light anesthesia to visualize the airways, sample secretions, or biopsy masses when you suspect malignancy, infection, or a foreign body
- Bronchoscopy is also a key tool in significant hemoptysis, because it can localize the bleeding source and allow therapeutic maneuvers such as cauterizing an endobronchial lesion, removing a mass, placing an endobronchial blocker, or even instilling glue to tamponade bleeding before the patient aspirates blood and asphyxiates

PFT Indications

Evaluation of symptoms such as chronic persistent cough, wheezing, dyspnea, and exertional cough or chest pain

Objective assessment of bronchodilator therapy

Evaluation of effects of exposure to dusts or chemicals at work

Risk evaluation of patients prior to thoracic or upper abdominal surgery

Objective assessment of respiratory impairment

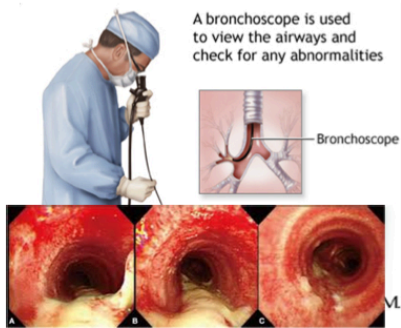
Monitoring disease course and response to therapy



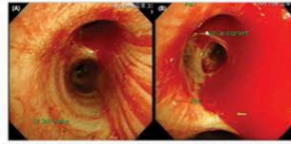
Pulmonary function tests are indicated to objectively evaluate chronic cough, wheeze, dyspnea, or exertional chest symptoms, to assess bronchodilator responsiveness to agents like albuterol, ipratropium, or levalbuterol, to monitor effects of occupational exposures in workers such as firefighters and police, to stratify perioperative risk before thoracic or upper abdominal surgery, and to quantify respiratory impairment in suspected neuromuscular disease or in patients after starting therapy

- PFTs also establish a new baseline after treatment so you can follow disease course and therapeutic response over time

Bronchoscopy



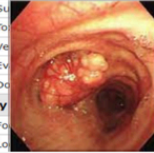
Pneumonia



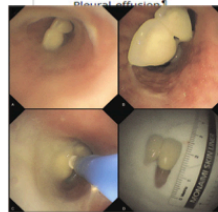
copy

Hemoptysis

Suspected tracheo-esophageal fistula
Chest trauma
Suspected tracheo-esophageal fistula
To
Ver
Ev
Do
May
For
Lo
Atelectasis (persistent)
Abnormal chest radiograph
Pleural effusion



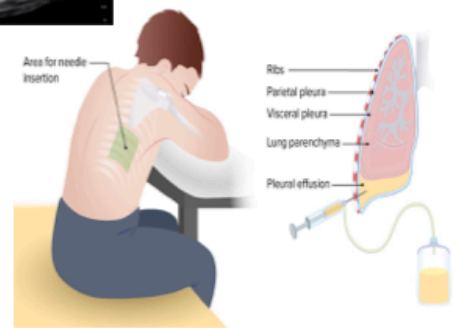
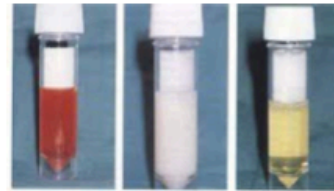
Endobronchial Tumor



or other procedure
or hyperlucency
fistula

Foreign mass
Pneumonia
(immunocompetent host)
Foreign Body
or suspected)
transplant recipient

Thoracentesis



Thoracentesis involves inserting a needle through the chest wall into the pleural space, guided by imaging, to sample pleural effusion fluid, and the gross appearance can suggest etiologies such as malignant, infectious, urinothorax, or chylous effusions (???)

- Pleural fluid must then be analyzed with laboratory studies and Light's criteria, comparing pleural and serum protein and LDH, to classify the effusion as transudative, which is often due to systemic conditions like CHF, hepatic hydrothorax, nephrotic syndrome, or urinothorax, versus exudative, which usually reflects local infection, malignancy, or inflammatory disease and requires more targeted interventions

Common Causes of Effusions

Transudative

- CHF
- Hepatic hydrothorax
- Nephrotic syndrome
- Urinothorax

Exudative

- Infections
- Malignancy
- Inflammatory conditions

Exudative Criteria

Compare Serum to Pleural Fluid

LIGHT'S CRITERIA		
	TRANSUDATIVE	EXUDATIVE
MECHANISM	↑ Capillary hydrostatic pressure ↓ Capillary oncotic pressure	↑ Capillary permeability
$\frac{\text{FLUID PROTEIN}}{\text{SERUM PROTEIN}}$	≤ 0.5	> 0.5
$\frac{\text{FLUID LDH}}{\text{SERUM LDH}}$	≤ 0.6	> 0.6
LDH	$\leq \frac{2}{3}$ the upper limit of normal serum LDH	$> \frac{2}{3}$ the upper limit of normal serum LDH
COMMON CAUSES	HEART FAILURE CIRRHOSIS NEPHROTIC SYNDROME	INFECTION MALIGNANCY PULMONARY EMBOLISM AUTOIMMUNE

Pleural Fluid Only 3 Test Combo (PFO3); + if any are positive

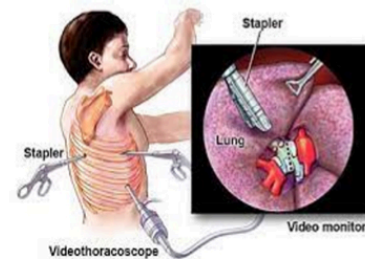
- Pleural fluid protein greater than 3.0 g/dL (30 g/L)
- Pleural fluid cholesterol greater than 55 mg/dL (1.424 mmol/L)
- Pleural fluid LDH greater than 0.67 times the upper limit of the laboratory's normal serum LDH

Lung Biopsy

Bronchoscopy

CT guided with
interventional radiology

Open lung via VATs or
thoracotomy



Lung biopsies can be obtained either from outside the chest or from within the airways, and the approach is chosen based on lesion location, patient risk, and need for tissue

- CT guided lung biopsy is performed by interventional radiology, where a needle is advanced through the chest wall into a radiographically localized lesion, often with the patient awake but lightly sedated
- Video assisted thoracoscopic surgery, or VATS, is done by cardiothoracic surgeons who place ports into the thorax under general anesthesia, allowing direct visualization and wedge resection or larger biopsies of lung tissue
- Bronchoscopic biopsies are obtained from inside the airway using a bronchoscope, and patients typically receive sedation but can remain awake in high risk cases where general anesthesia is unsafe

Radiologicraphic Imaging
Multiple Dr. Rachlin Lectures

CXR

- Various Positions – PA, Lateral, Decubitus

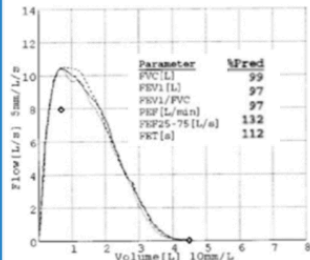
CT Scan

- With or Without Contrast
- High-Resolution

Ultrasound

Case Studies

Case 1



mild tachypnea, flushed appearance, Temp 102.3 F, clear lungs with equal/good air entry

Spirometry: normal

Case 2

58-year-old, previous smoker.



Spirometry: Moderate Restriction

CT Scan: ground glass opacities, honeycomb pattern - predominantly in subpleural zones, traction bronchiectasis.

Case 3



CXR: Hyperinflation of bilateral lung fields

Case Studies

Case 1

23-year-old college student, previously healthy, presents with fever for 4 days, associated body aches, fatigue and non-productive cough.

**ACUTE INFLUENZA
INFECTION**

Case 2

58-year-old former smoker who quit 10 years ago, presents with three weeks of cough with scarce yellowish sputum, severe exertional dyspnea and weakness.

**IDIOPATHIC
PULMONARY FIBROSIS**

Case 3

13-year-old male, known history of severe eczema and environmental allergies, presents with progressive exertional dyspnea, wheeze, and nocturnal cough. Symptoms improve with rest.

**PERSISTENT POORLY
CONTROLLED ASTHMA**

OVERVIEW



Taking a medical history of patients with lung disease



Pathophysiology of Presenting Symptoms



Technique & Interpretation of the lung and thorax PE



Basic pulmonary diagnostic techniques



Understand High Risk behavior for lung disease

Review Vaccine Preventable Respiratory Illness

Review additional Preventable Diseases

Key Facts about Smoking



Smoking is the #1 cause of preventable disease and death worldwide

Smoking-related diseases claim more than 480,000 lives in the U.S. each year

This is nearly 1 in 5 deaths

Smoking costs the U.S. at least \$600 billion each year

 > \$372 billion in lost productivity & \$260 billion in direct healthcare expenditures

In 2022, ~ 12 of every 100 U.S. adults $\geq$ 18 years (11.6%) currently smoked cigarettes

Among adults who have ever smoked daily, 78% had smoked their first cigarette by the time they were 18 years of age, and 94% had by age 21

 $\frac{1}{2}$ of these smokers will die prematurely, 10 years before their never-smoking peers

Key Facts about Smoking



2021

Cigarette smoking is down, but about
34 MILLION
American adults still smoke



Cigarette smoking remains high
among certain groups



Men



Adults 25-64
years old



Lower education



Below
poverty level



Midwest
and South



Uninsured
or Medicaid



Disabled



Serious
psychological
distress



American Indians,
Alaska Natives and
Multiracial



Lesbians, gays,
and bisexuals

Strategies essential to continue reducing
cigarette smoking overall



Implement
smoke-free laws



Run mass
media campaigns



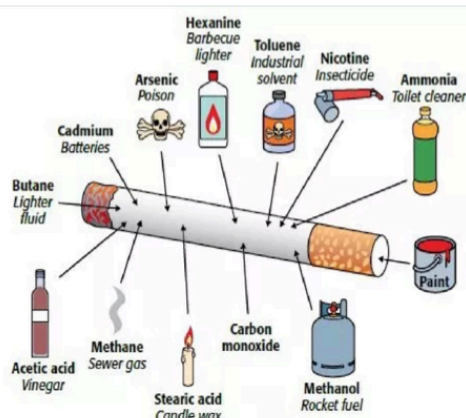
Raise
tobacco prices



Make quit help
easy to access

https://www.cdc.gov/tobacco/data_statistics/fact_sheets/fast_facts/cigarette-smoking-in-the-us.html

Contents of a Cigarette



Cigarette smoke contains > 7,000 chemicals,
at least 69 of which are known to cause cancer



U.S. Department of Health and Human Services. How Tobacco Smoke Causes Disease:
The Biology and Behavioral Basis for Smoking-Attributable Disease A Report of the Surgeon General. 2010

Tobacco Use Risks

Smoking causes about 90% of all lung cancer deaths. More women die from lung cancer each year than from breast cancer

Smoking causes about 80% of all deaths from chronic obstructive pulmonary disease (COPD)

Cigarette smoking increases risk for death from all causes in men and women.

The risk of dying from cigarette smoking has increased over the last 50 years in the U.S.

Estimates show smoking increases the risk:

- For coronary heart disease by 2 to 4 times
- For stroke by 2 to 4 times



U.S. Department of Health and Human Services. [The Health Consequences of Smoking—50 Years of Progress: A Report of the Surgeon General](#). Atlanta: U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, National Center for Chronic Disease Prevention and Health Promotion, Office on Smoking and Health, 2014 [accessed 2017 Apr 20].

Tobacco Use Risks

Smoking can cause cancer almost anywhere in your body:

If nobody smoked, one of every three cancer deaths in the United States would not happen

Bladder
Blood (acute myeloid leukemia)
Cervix
Colon and rectum (colorectal)
Esophagus
Kidney and ureter
Larynx
Liver
Oropharynx
Pancreas
Stomach
Trachea, bronchus, and lung

Risks from Smoking



U.S. Department of Health and Human Services. [How Tobacco Smoke Causes Disease: What It Means to You](#). Atlanta: U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, National Center for Chronic Disease Prevention and Health Promotion, Office on Smoking and Health, 2010 [accessed 2017 Apr 20].

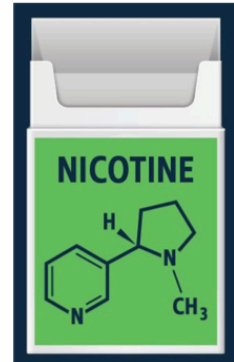
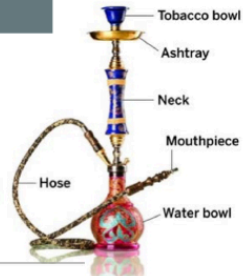
Tobacco Products

Cigarettes

Electronic Cigarette/Vaping

Hookah

Pouches



US market since 2006



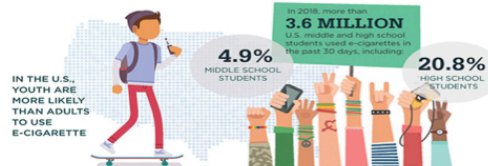
Vaping / E-Cigarette

Vaping is the act of inhaling and exhaling an aerosol, often referred to as vapor, produced by an e-cigarette, vape pen, or similar device



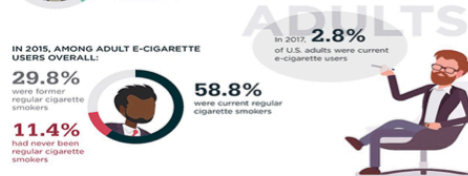
It is difficult for consumers to know what e-cigarette products contain. For example, some e-cigarettes marketed as containing zero percent nicotine have been found to contain nicotine.

E-CIGARETTES ARE THE MOST COMMONLY USED TOBACCO PRODUCT AMONG YOUTH.



AMONG CURRENT E-CIGARETTE USERS AGED 45 YEARS AND OLDER in 2015, most were either current or former regular cigarette smokers, and 1.3% had never been cigarette smokers.

BY CONTRAST, AMONG CURRENT E-CIGARETTE USERS AGED 18-24 YEARS, **40.0%** had NEVER BEEN regular cigarette smokers



Use of Tobacco Products Starts Young

7.2% of all students (2.01 million) reported current use of tobacco products

9.5% of high school students (1.50 million) reported current use of tobacco products

4.1% of middle school students (480,000) reported current use of tobacco products

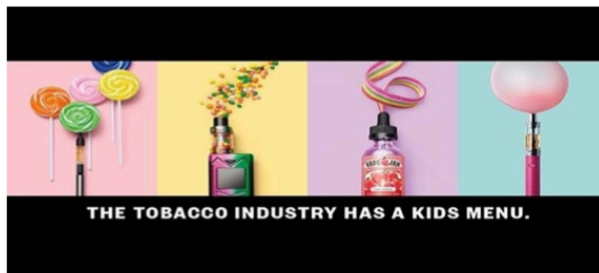


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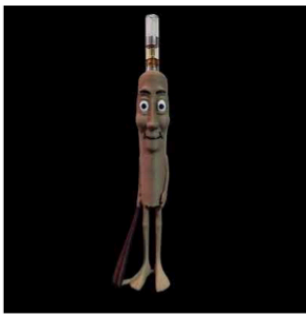
4.1% of middle school students (480,000) reported current use of tobacco products



Kids are lured in by these products



- Kids are appealed to fun, bright, and trendy things; and when vapes advertise seemingly yummy flavors, with trendy holders, and new technology, children are incredibly susceptible to addiction



Triple T Vape Holder



Interactive Technology on Vape



"Yummy" Flavors



Do kids really Vape?

Reasons Kids Vape

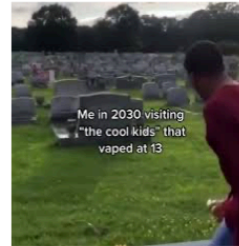
- To “fit in” and “be cool”
- Curious or bored
- Social media (post a picture)
- Don’t understand that it is harmful
- Continue to vape because they are addicted

Facts about Vaping in Kids

4 times increased risk of smoking regular cigarettes

Increased risk of developing a nicotine addiction

Increased risk of developing addiction to other drugs of abuse



Facts About Vaping in Kids

- 7.1% of high school students (1.13 million) reported current use of e-cigarettes
- 2.6% of middle school students (300,000) reported current use of e-cigarettes

Of these kids, 43% of ever users are current users



Tobacco product use among middle and high school students in the United States: National Youth Tobacco Survey, 2025

Nicotine & Tobacco Research, 2026, 00, 1-9
<https://doi.org/10.1093/ntn/ntag116>
Published: 23 June 2026
Original Investigation

Facts About Vaping in Kids

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- 2.6% of middle school students (300,000) reported current use of e-cigarettes

Of these kids, 43% of ever users are current users

67% of those kids have tried to quit



Tobacco product use among middle and high school students in the United States: National Youth Tobacco Survey, 2025

Nicotine & Tobacco Research, 2026, 00, 1-9
<https://doi.org/10.1093/ntn/ntag116>
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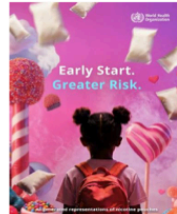
Nicotine Pouches

- **Current Use**
- 1.7% of students (460,000) reported current use of nicotine pouches
- **Frequency of Use**
- More than 1 in 6 (17.6%) youth nicotine pouch users reported using nicotine pouches daily
- More than 1 in 4 (26.3%) youth nicotine pouch users reported using nicotine pouches on at least 20 of the last 30 days
- **Flavored Nicotine Pouch Use**
- More than 9 out of 10 (90.8%) current nicotine pouch users used flavored nicotine pouches, with mint flavors being the most popular, followed by fruit



Nicotine Pouches

- **Surging Use:** use among youth and young adults nearly **quadrupled** between 2022 and 2025
- **Second Most Popular:** Pouches have become the second most commonly used tobacco/nicotine product among middle and high school students, trailing only e-cigarettes
- **Social Media Influence:** Online trends on platforms like TikTok—promoted by so-called "Zynfluencers"—frequently market the product as a lifestyle accessory linked to focus, energy, and masculinity
- **Discreet Appeal:** Teens and young adults favor pouches because they are small, odorless, spitless, and easily hidden from teachers and parents
- **Health and Addiction Concerns:** Health organizations like the World Health Organization (WHO) and the CDC warn that high nicotine concentrations in these pouches lead to rapid dependence.



Health Risks in Children



Nicotine can harm adolescent brain development, which continues into the early to mid-20s



Using nicotine in adolescence can harm the parts of the brain that control attention, learning, mood, and impulse control



Using nicotine in adolescence may also increase risk for future addiction to other drugs



The use of all tobacco products is unsafe for kids, teens, young adults, and adults

Hookah

A hookah is a water pipe.

The hookah pipe tobacco has the same chemicals and is as unsafe as cigarettes

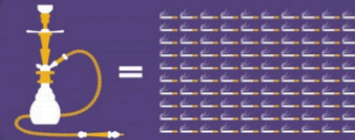
The water in the hookah does not filter out the toxic ingredients in the tobacco smoke

One session of hookah pipe smoking lasts about 45 minutes to an hour (while a cigarette lasts just a few minutes)

During that time hookah users may breathe in as much smoke as a cigarette smoker would from 100 or more cigarettes

That's 5 packs of cigarettes in one session!

Smoking a hookah is the equivalent of smoking **100 cigarettes**

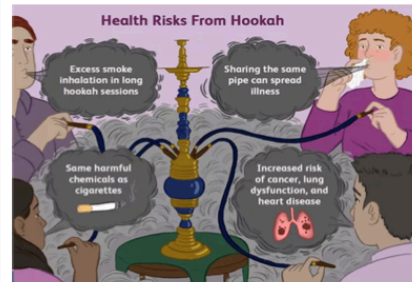


Source: World Health Organization (2005), mentioned in the October 2010 issue of The Journal of the American Dental Association

Learn more at MouthHealthy.org



Health Risks From Hookah



QUIT SMOKING? THIS IS HOW THE BODY CHANGES AFTER THE FINAL CIGARETTE



-20 minutes

Pulse rate returns to normal

-8 hours

Nicotine and carbon monoxide levels in blood reduce by more than half and oxygen levels return to normal

-48 hours

Carbon monoxide will be eliminated from the body. Lungs start to clear out mucus and other smoking debris. There is no more nicotine in the body. The ability to taste and smell is improved.

-72 hours

Breathing becomes easier. Bronchial tubes begin to relax and energy levels increase.

-12 weeks

Circulation is improved

-9 months

Coughs, wheezing and breathing problems improve as lung function increases by up to 10 per cent.

-1 year

Risk of heart disease is about half compared with a person who's still smoking.

-15 years

Risk of heart attack falls to the same as someone who has never smoked.

Smoking: Timeline of how giving up cigarettes affects the body (Image: EXPRESSIONS)

lifecoachtoquitsmoking.com

Benefits of Quitting

Just **1 year** after quitting smoking, your **risk for a heart attack drops sharply**

Within **2 to 5 years** after quitting smoking, your **risk for stroke may reduce to about that of a nonsmoker's**

If you quit smoking, your **risks for cancers of the mouth, throat, esophagus, and bladder drop by half within 5 years**

Ten years after you quit smoking, your risk for dying from **lung cancer drops by half**

US Adult Smoking Trends in 2025:

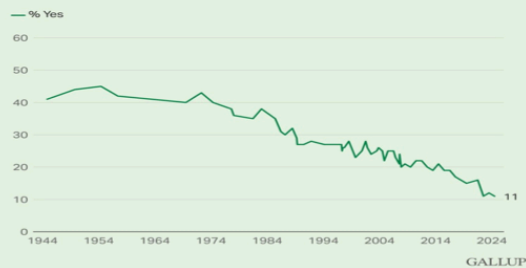
Smoking is on the decline

Cigarette smoking: 9.1% of adults aged ≥18 report current use

E-cigarette use: 6.7% of adults use e-cigarettes, up from earlier years but still below cigarette prevalence

Cigarette Smoking Among U.S. Adults at 80-Year Low

Have you, yourself, smoked any cigarettes in the past week?



U.S. Adult Smoking Decline

Long-term trend: From 42.4% of adults smoking in 1965 to **11.6% in 2022**, a **72.6% drop** over 57 years

Recent pace: Over the past five years, adult smoking rates fell 17 percentage points, from 14.0% in 2017 to 11.6% in 2022

By sex: Men's rates have fallen faster (from 51.9% in 1965 to 13.2% in 2022) than women's (from 33.9% to 10.0%)

By race: Non-Hispanic White adults saw the largest share decline (from 42.1% to 12.7%), while American Indian/Alaska Native adults have been more stable

By age: Smoking is highest among adults aged 45–64 (14.9% in 2022), with the steepest declines in younger adults



US Adult Smoking Trends in 2025: Smoking is on the decline



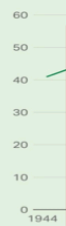
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Cigarette Smoking Among U.S. Adults at 80-Year Low

Have you, yourself, smoked any cigarettes in the past week?

— % Yes



WHO 2025 report:

1 in 5 adults worldwide still use tobacco, with prevalence among men (32.5% in 2024), still higher than women (6.6%)

U.S. Adult Smoking Decline

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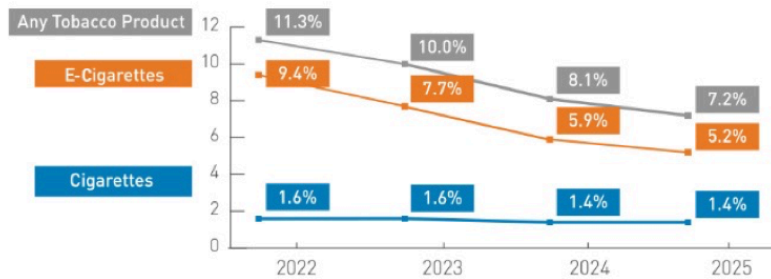
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From 2022-2025 *YOUTH TOBACCO USE DECLINED*



Source: National Youth Tobacco Survey 2022-2025



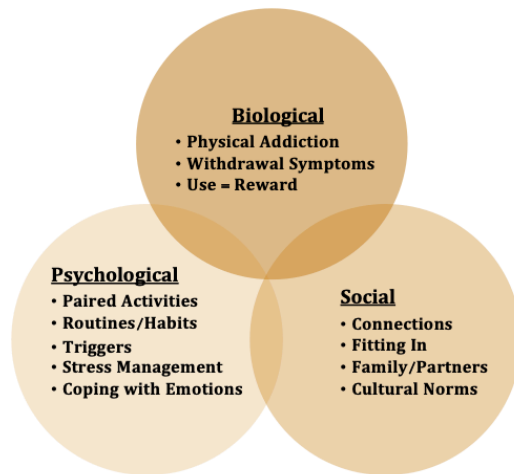
Smoking Cessation

70% of Smokers are Interested in Quitting

50% Make an Attempt to Quit

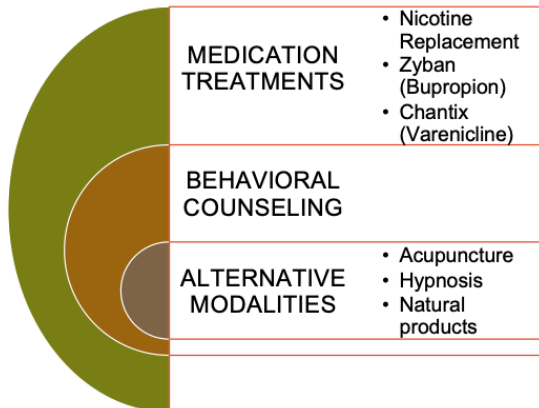
Average of 30 quit attempts for successful cessation

4-7% Succeed



Glynn et al. (2018)
Chaitonet al. (2016)
Institute for health and recovery

Evidence Based Tobacco Cessation Treatments



NRT

The main goal is to reduce the nicotine withdrawal/ craving and ultimately cut back and quit

OTC: patch, gum, lozenge

Rx: inhaler (off market), nasal spray

Zyban (Bupropion)

Quit date is 1 week **prior** to starting medication
Duration is 12 wk typically
May cause weight loss

Chantix (Varenicline)

Reduces nicotine withdrawal symptoms and reduces the rewarding aspects of smoking/reduces want to smoke.

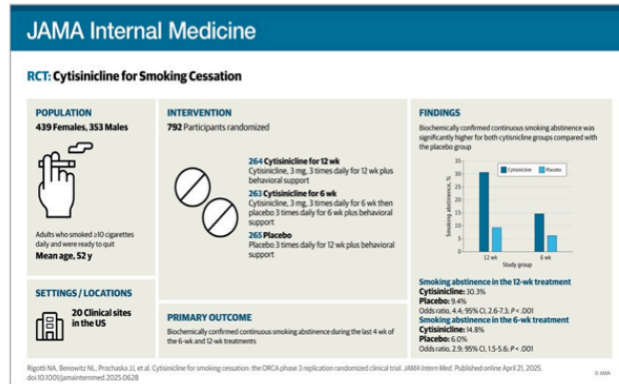
Stop smoking 1-5 wk after initiating

Take for a 12 wk course
If you are successful can continued another 12 wk to prevent relapse.

What About That New Drug?

CYTISINICLINE

- binds to the same receptors as Chantix so has similar properties of reducing cravings and withdrawal symptoms
- as of 6/2025, it is under review as a NDA(new drug application) to the FDA
- breakthrough therapy designation for vaping cessation (may be 1st approved drug for this)
- promising results in Phase III trials for both smoking and vaping cessation



CYTISINICLINE

Approval Timeline and Status



FDA Review Delay: The U.S. Food and Drug Administration (FDA) issued a Complete Response Letter (CRL) in June 2026, deferring approval due to general manufacturing facility issues from a third-party partner rather than any safety or effectiveness concerns



Resubmission Plans: Achieve Life Sciences plans to resubmit its New Drug Application in the fourth quarter of 2026 using a new primary manufacturing partner, Adare Pharma Solutions



Expected Availability: If the resubmission goes well, approval and a subsequent U.S. commercial launch could happen during the first half of 2027